Announcement: pandas 3.0 released! Read more in the blog post ✕

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Working with missing data

Values considered "missing"

pandas uses different sentinel values to represent a missing (also referred to as NA) depending on the data type.

`numpy.nan` for NumPy data types. The disadvantage of using NumPy data types is that the original data type will be coerced to `np.float64` or `object`.

```
In [1]: pd.Series([1, 2], dtype=np.int64).reindex([0, 1, 2])
Out[1]:
0    1.0
1    2.0
2    NaN
dtype: float64

In [2]: pd.Series([True, False], dtype=np.bool_).reindex([0, 1, 2])
Out[2]:
0     True
1    False
2      NaN
dtype: object
```

`NaT` for NumPy `np.datetime64`, `np.timedelta64`, and `PeriodDtype`. For typing applications, use `api.typing.NaTType`.

```
In [3]: pd.Series([1, 2], dtype=np.dtype("timedelta64[ns]")).reindex([0, 1, 2])
Out[3]:
0   0 days 00:00:00.000000001
1   0 days 00:00:00.000000002
2                         NaT
dtype: timedelta64[ns]

In [4]: pd.Series([1, 2], dtype=np.dtype("datetime64[ns]")).reindex([0, 1, 2])
Out[4]:
0   1970-01-01 00:00:00.000000001
1   1970-01-01 00:00:00.000000002
2                             NaT
dtype: datetime64[ns]

In [5]: pd.Series(["2020", "2020"], dtype=pd.PeriodDtype("D")).reindex([0, 1, 2])
Out[5]:
0    2020-01-01
1    2020-01-01
2           NaT
dtype: period[D]
```

`NA` for `StringDtype`, `Int64Dtype` (and other bit widths), `Float64Dtype` (and other bit widths), `BooleanDtype` and `ArrowDtype`. These types will maintain the original data type of the data. For typing applications, use `api.typing.NAType`.

```
In [6]: pd.Series([1, 2], dtype="Int64").reindex([0, 1, 2])
Out[6]:
0       1
1       2
2    <NA>
dtype: Int64

In [7]: pd.Series([True, False], dtype="boolean[pyarrow]").reindex([0, 1, 2])
Out[7]:
0     True
1    False
2     <NA>
dtype: bool[pyarrow]
```

To detect these missing value, use the `isna()` or `notna()` methods.

```
In [8]: ser = pd.Series([pd.Timestamp("2020-01-01"), pd.NaT])

In [9]: ser
Out[9]:
0   2020-01-01
1          NaT
dtype: datetime64[us]

In [10]: pd.isna(ser)
Out[10]:
0    False
1     True
dtype: bool
```

Note

`isna()` or `notna()` will also consider `None` a missing value.

```
In [11]: ser = pd.Series([1, None], dtype=object)

In [12]: ser
Out[12]:
0       1
1    None
dtype: object

In [13]: pd.isna(ser)
Out[13]:
0    False
1     True
dtype: bool
```

Warning

Equality comparisons between `np.nan`, `NaT`, and `NA` do not act like `None`

```
In [14]: None == None  # noqa: E711
Out[14]: True

In [15]: np.nan == np.nan
Out[15]: False

In [16]: pd.NaT == pd.NaT
Out[16]: False

In [17]: pd.NA == pd.NA
Out[17]: <NA>
```

Therefore, an equality comparison between a `DataFrame` or `Series` with one of these missing values does not provide the same information as `isna()` or `notna()`.

```
In [18]: ser = pd.Series([True, None], dtype="boolean[pyarrow]")

In [19]: ser == pd.NA
Out[19]:
0    <NA>
1    <NA>
dtype: bool[pyarrow]

In [20]: pd.isna(ser)
Out[20]:
0    False
1     True
dtype: bool
```

`NA` semantics

Warning

Experimental: the behaviour of `NA` can still change without warning.

Starting from pandas 1.0, an experimental `NA` value (singleton) is available to represent scalar missing values. The goal of `NA` is provide a "missing" indicator that can be used consistently across data types (instead of `np.nan`, `None` or `pd.NaT` depending on the data type).

For example, when having missing values in a `Series` with the nullable integer dtype, it will use `NA`:

```
In [21]: s = pd.Series([1, 2, None], dtype="Int64")

In [22]: s
Out[22]:
0       1
1       2
2    <NA>
dtype: Int64

In [23]: s[2]
Out[23]: <NA>

In [24]: s[2] is pd.NA
Out[24]: True
```

Currently, pandas does not use those data types using `NA` by default in a `DataFrame` or `Series`, so you need to specify the dtype explicitly. An easy way to convert to those dtypes is explained in the conversion section.

Propagation in arithmetic and comparison operations

In general, missing values *propagate* in operations involving `NA`. When one of the operands is unknown, the outcome of the operation is also unknown.

For example, `NA` propagates in arithmetic operations, similarly to `np.nan`:

```
In [25]: pd.NA + 1
Out[25]: <NA>

In [26]: "a" * pd.NA
Out[26]: <NA>
```

There are a few special cases when the result is known, even when one of the operands is `NA`.

```
In [27]: pd.NA ** 0
Out[27]: 1

In [28]: 1 ** pd.NA
Out[28]: 1
```

In equality and comparison operations, `NA` also propagates. This deviates from the behaviour of `np.nan`, where comparisons with `np.nan` always return `False`.

```
In [29]: pd.NA == 1
Out[29]: <NA>

In [30]: pd.NA == pd.NA
Out[30]: <NA>

In [31]: pd.NA < 2.5
Out[31]: <NA>
```

To check if a value is equal to `NA`, use `isna()`

```
In [32]: pd.isna(pd.NA)
Out[32]: True
```

Note

An exception on this basic propagation rule are *reductions* (such as the mean or the minimum), where pandas defaults to skipping missing values. See the calculation section for more.

Logical operations

For logical operations, `NA` follows the rules of the three-valued logic (or *Kleene logic*, similarly to R, SQL and Julia). This logic means to only propagate missing values when it is logically required.

For example, for the logical "or" operation (`|`), if one of the operands is `True`, we already know the result will be `True`, regardless of the other value (so regardless the missing value would be `True` or `False`). In this case, `NA` does not propagate:

```
In [33]: True | False
Out[33]: True

In [34]: True | pd.NA
Out[34]: True

In [35]: pd.NA | True
Out[35]: True
```

On the other hand, if one of the operands is `False`, the result depends on the value of the other operand. Therefore, in this case `NA` propagates:

```
In [36]: False | True
Out[36]: True

In [37]: False | False
Out[37]: False

In [38]: False | pd.NA
Out[38]: <NA>
```

The behaviour of the logical "and" operation (`&`) can be derived using similar logic (where now `NA` will not propagate if one of the operands is already `False`):

```
In [39]: False & True
Out[39]: False

In [40]: False & False
Out[40]: False

In [41]: False & pd.NA
Out[41]: False
```

```
In [42]: True & True
Out[42]: True

In [43]: True & False
Out[43]: False

In [44]: True & pd.NA
Out[44]: <NA>
```

`NA` in a boolean context

Since the actual value of an NA is unknown, it is ambiguous to convert NA to a boolean value.

```
In [45]: bool(pd.NA)
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[45], line 1
----> 1 bool(pd.NA)

File pandas/_libs/missing.pyx:415, in pandas._libs.missing.NAType.__bool__()
--> 415 'Could not get source, probably due dynamically evaluated source code.'
TypeError: boolean value of NA is ambiguous
```

This also means that `NA` cannot be used in a context where it is evaluated to a boolean, such as `if condition: ...` where `condition` can potentially be `NA`. In such cases, `isna()` can be used to check for `NA` or `condition` being `NA` can be avoided, for example by filling missing values beforehand.

A similar situation occurs when using `Series` or `DataFrame` objects in `if` statements, see Using if/truth statements with pandas.

NumPy ufuncs

`pandas.NA` implements NumPy's `__array_ufunc__` protocol. Most ufuncs work with `NA`, and generally return `NA`:

```
In [46]: np.log(pd.NA)
Out[46]: <NA>

In [47]: np.add(pd.NA, 1)
Out[47]: <NA>
```

Warning

Currently, ufuncs involving an ndarray and `NA` will return an object-dtype filled with NA values.

```
In [48]: a = np.array([1, 2, 3])

In [49]: np.greater(a, pd.NA)
Out[49]: array([<NA>, <NA>, <NA>], dtype=object)
```

The return type here may change to return a different array type in the future.

See DataFrame interoperability with NumPy functions for more on ufuncs.

Conversion

If you have a `DataFrame` or `Series` using `np.nan`, `DataFrame.convert_dtypes()` and `Series.convert_dtypes()`, respectively, will convert your data to use the nullable data types supporting `NA`, such as `Int64Dtype` or `ArrowDtype`. This is especially helpful after reading in data sets from IO methods where data types were inferred.

```
In [50]: import io

In [51]: data = io.StringIO("a,b\n,True\n2,")

In [52]: df = pd.read_csv(data)

In [53]: df.dtypes
Out[53]:
a    float64
b     object
dtype: object

In [54]: df_conv = df.convert_dtypes()

In [55]: df_conv
Out[55]:
      a     b
0  <NA>  True
1     2  <NA>

In [56]: df_conv.dtypes
Out[56]:
a      Int64
b    boolean
dtype: object
```

Inserting missing data

You can insert missing values by simply assigning to a `Series` or `DataFrame`. The missing value sentinel used will be chosen based on the dtype.

```
In [57]: ser = pd.Series([1., 2., 3.])

In [58]: ser.loc[0] = None

In [59]: ser
Out[59]:
0    NaN
1    2.0
2    3.0
dtype: float64

In [60]: ser = pd.Series([pd.Timestamp("2021"), pd.Timestamp("2021")])

In [61]: ser.iloc[0] = np.nan

In [62]: ser
Out[62]:
0          NaT
1   2021-01-01
dtype: datetime64[us]

In [63]: ser = pd.Series([True, False], dtype="boolean[pyarrow]")

In [64]: ser.iloc[0] = None

In [65]: ser
Out[65]:
0     <NA>
1    False
dtype: bool[pyarrow]
```

For `object` types, pandas will use the value given:

```
In [66]: s = pd.Series(["a", "b", "c"], dtype=object)

In [67]: s.loc[0] = None

In [68]: s.loc[1] = np.nan

In [69]: s
Out[69]:
0    None
1     NaN
2       c
dtype: object
```

Calculations with missing data

Missing values propagate through arithmetic operations between pandas objects.

```
In [70]: ser1 = pd.Series([np.nan, np.nan, 2, 3])

In [71]: ser2 = pd.Series([np.nan, 1, np.nan, 4])

In [72]: ser1
Out[72]:
0    NaN
1    NaN
2    2.0
3    3.0
dtype: float64

In [73]: ser2
Out[73]:
0    NaN
1    1.0
2    NaN
3    4.0
dtype: float64

In [74]: ser1 + ser2
Out[74]:
0    NaN
1    NaN
2    NaN
3    7.0
dtype: float64
```

The descriptive statistics and computational methods discussed in the data structure overview (and listed here and here) all account for missing data.

When summing data, NA values or empty data will be treated as zero.

```
In [75]: pd.Series([np.nan]).sum()
Out[75]: np.float64(0.0)

In [76]: pd.Series([], dtype="float64").sum()
Out[76]: np.float64(0.0)
```

When taking the product, NA values or empty data will be treated as 1.

```
In [77]: pd.Series([np.nan]).prod()
Out[77]: np.float64(1.0)

In [78]: pd.Series([], dtype="float64").prod()
Out[78]: np.float64(1.0)
```

Cumulative methods like `cumsum()` and `cumprod()` ignore NA values by default, but preserve them in the resulting array. To override this behaviour and include NA values in the calculation, use `skipna=False`.

```
In [79]: ser = pd.Series([1, np.nan, 3, np.nan])

In [80]: ser
Out[80]:
0    1.0
1    NaN
2    3.0
3    NaN
dtype: float64

In [81]: ser.cumsum()
Out[81]:
0    1.0
1    NaN
2    4.0
3    NaN
dtype: float64

In [82]: ser.cumsum(skipna=False)
Out[82]:
0    1.0
1    NaN
2    NaN
3    NaN
dtype: float64
```

Dropping missing data

`dropna()` drops rows or columns with missing data.

```
In [83]: df = pd.DataFrame([[np.nan, 1, 2], [1, 2, np.nan], [1, 2, 3]])

In [84]: df
Out[84]:
     0  1    2
0  NaN  1  2.0
1  1.0  2  NaN
2  1.0  2  3.0

In [85]: df.dropna()
Out[85]:
     0  1    2
2  1.0  2  3.0

In [86]: df.dropna(axis=1)
Out[86]:
   1
0  1
1  2
2  2

In [87]: ser = pd.Series([1, pd.NA], dtype="int64[pyarrow]")

In [88]: ser.dropna()
Out[88]:
0    1
dtype: int64[pyarrow]
```

Filling missing data

Filling by value

`fillna()` replaces NA values with non-NA data.

Replace NA with a scalar value

```
In [89]: data = {"np": [1.0, np.nan, np.nan, 2], "arrow": pd.array([1.0, pd.NA, pd.

In [90]: df = pd.DataFrame(data)

In [91]: df
Out[91]:
    np  arrow
0  1.0    1.0
1  NaN   <NA>
2  NaN   <NA>
3  2.0    2.0

In [92]: df.fillna(0)
Out[92]:
    np  arrow
0  1.0    1.0
1  0.0    0.0
2  0.0    0.0
3  2.0    2.0
```

When the data has object dtype, you can control what type of NA values are present.

```
In [93]: df = pd.DataFrame({"a": [pd.NA, np.nan, None]}, dtype=object)

In [94]: df
Out[94]:
      a
0  <NA>
1   NaN
2  None

In [95]: df.fillna(None)
Out[95]:
      a
0  None
1  None
2  None

In [96]: df.fillna(np.nan)
Out[96]:
     a
0  NaN
1  NaN
2  NaN

In [97]: df.fillna(pd.NA)
Out[97]:
      a
0  <NA>
1  <NA>
2  <NA>
```

However when the dtype is not object, these will all be replaced with the proper NA value for the dtype.

```
In [98]: data = {"np": [1.0, np.nan, np.nan, 2], "arrow": pd.array([1.0, pd.NA, pd.

In [99]: df = pd.DataFrame(data)

In [100]: df
Out[100]:
    np  arrow
0  1.0    1.0
1  NaN   <NA>
2  NaN   <NA>
3  2.0    2.0

In [101]: df.fillna(None)
Out[101]:
    np  arrow
0  1.0    1.0
1  NaN   <NA>
2  NaN   <NA>
3  2.0    2.0

In [102]: df.fillna(np.nan)
Out[102]:
    np  arrow
0  1.0    1.0
1  NaN   <NA>
2  NaN   <NA>
3  2.0    2.0

In [103]: df.fillna(pd.NA)
Out[103]:
    np  arrow
0  1.0    1.0
1  NaN   <NA>
2  NaN   <NA>
3  2.0    2.0
```

Fill gaps forward or backward

```
In [104]: df.ffill()
Out[104]:
    np  arrow
0  1.0    1.0
1  1.0    1.0
2  1.0    1.0
3  2.0    2.0

In [105]: df.bfill()
Out[105]:
    np  arrow
0  1.0    1.0
1  2.0    2.0
2  2.0    2.0
3  2.0    2.0
```

Limit the number of NA values filled

```
In [106]: df.ffill(limit=1)
Out[106]:
    np  arrow
0  1.0    1.0
1  1.0    1.0
2  NaN   <NA>
3  2.0    2.0
```

NA values can be replaced with corresponding value from a `Series` or `DataFrame` where the index and column aligns between the original object and the filled object.

```
In [107]: dff = pd.DataFrame(np.arange(30, dtype=np.float64).reshape(10, 3), column

In [108]: dff.iloc[3:5, 0] = np.nan

In [109]: dff.iloc[4:6, 1] = np.nan

In [110]: dff.iloc[5:8, 2] = np.nan

In [111]: dff
Out[111]:
      A     B     C
0   0.0   1.0   2.0
1   3.0   4.0   5.0
2   6.0   7.0   8.0
3   NaN  10.0  11.0
4   NaN   NaN  14.0
5  15.0   NaN   NaN
6  18.0  19.0   NaN
7  21.0  22.0   NaN
8  24.0  25.0  26.0
9  27.0  28.0  29.0

In [112]: dff.fillna(dff.mean())
Out[112]:
       A     B          C
0   0.00   1.0   2.000000
1   3.00   4.0   5.000000
2   6.00   7.0   8.000000
3  14.25  10.0  11.000000
4  14.25  14.5  14.000000
5  15.00  14.5  13.571429
6  18.00  19.0  13.571429
7  21.00  22.0  13.571429
8  24.00  25.0  26.000000
9  27.00  28.0  29.000000
```

Note

`DataFrame.where()` can also be used to fill NA values. Same result as above.

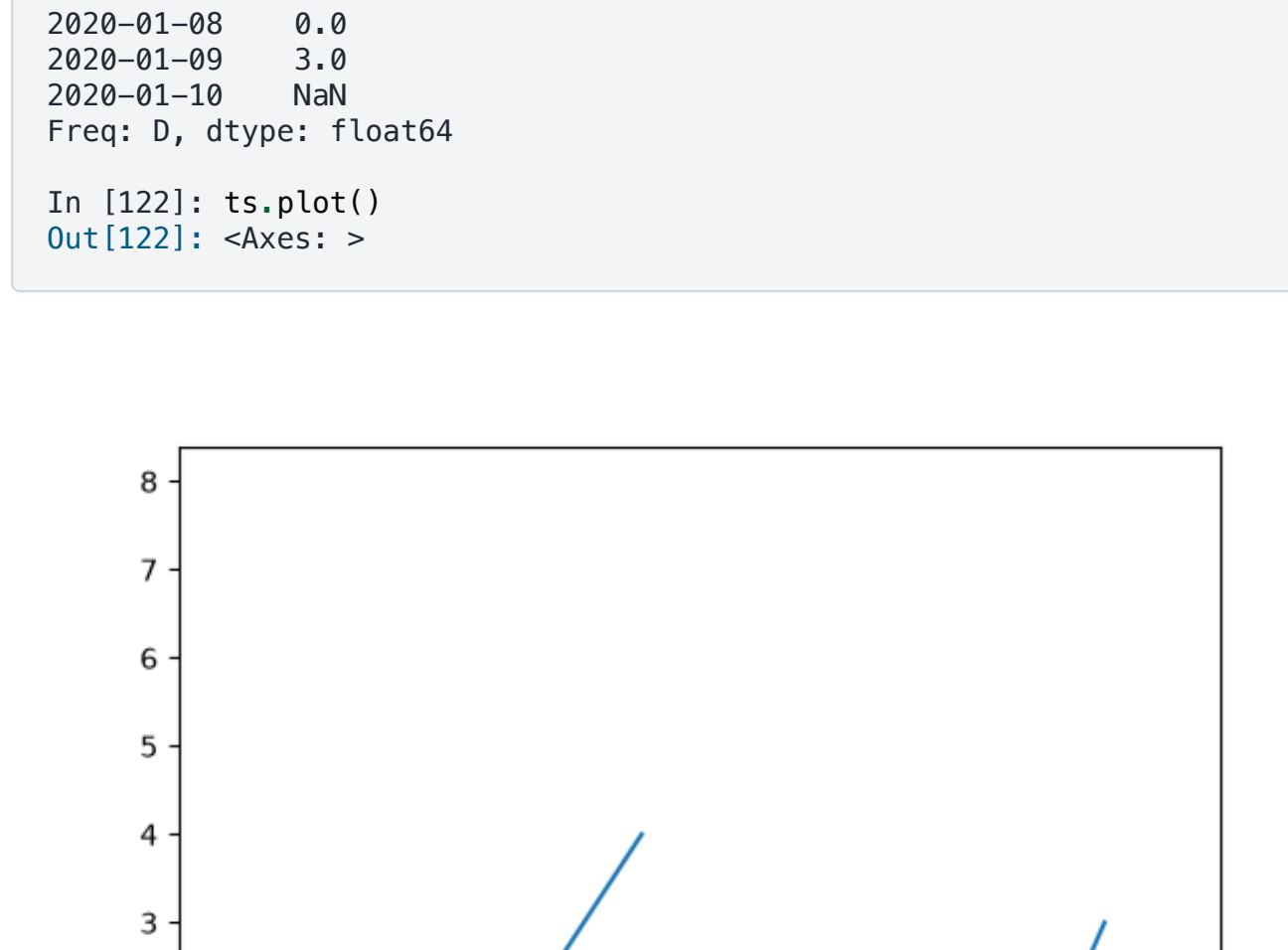
```
In [113]: dff.where(pd.notna(dff), dff.mean(), axis="columns")
Out[113]:
       A     B          C
0   0.00   1.0   2.000000
```

```
1   3.00   4.0   5.000000
2   6.00   7.0   8.000000
3  14.25  10.0  11.000000
4  14.25  14.5  14.000000
5  15.00  14.5  13.571429
6  18.00  19.0  13.571429
7  21.00  22.0  13.571429
8  24.00  25.0  26.000000
9  27.00  28.0  29.000000
```

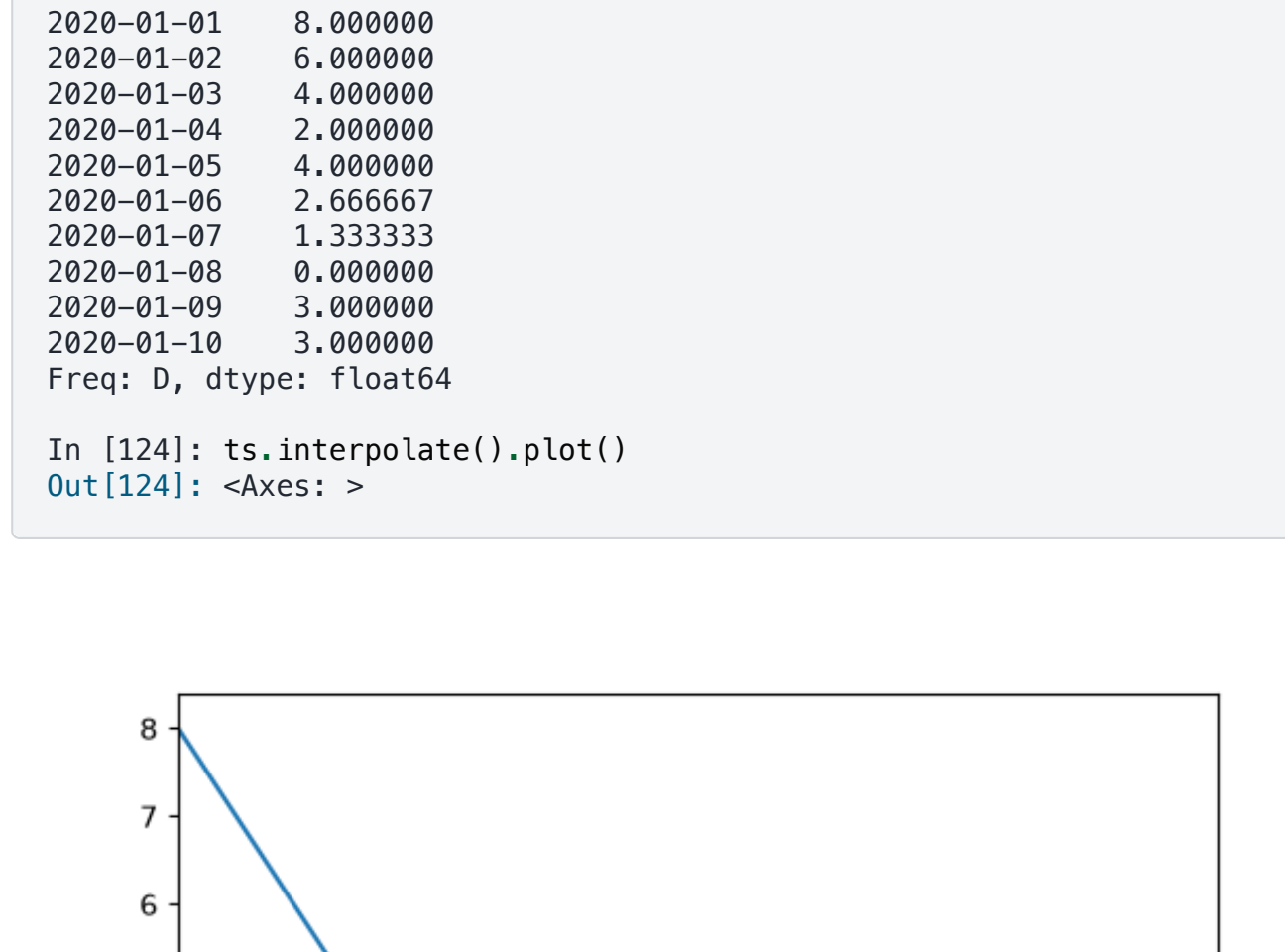
Interpolation

`DataFrame.interpolate()` and `Series.interpolate()` fills NA values using various interpolation methods.

```
In [114]: df = pd.DataFrame(
   .....:     {
   .....:         "A": [1, 2.1, np.nan, 4.7, 5.6, 6.8],
   .....:         "B": [0.25, np.nan, np.nan, 4, 12.2, 14.4],
   .....:     }
   .....: )
   .....:

In [115]: df
Out[115]:
     A      B
0  1.0   0.25
1  2.1    NaN
2  NaN    NaN
3  4.7   4.00
4  5.6  12.20
5  6.8  14.40

In [116]: df.interpolate()
Out[116]:
     A      B
0  1.0   0.25
1  2.1   1.50
2  3.4   2.75
3  4.7   4.00
4  5.6  12.20
5  6.8  14.40

In [117]: idx = pd.date_range("2020-01-01", periods=10, freq="D")

In [118]: data = np.random.default_rng(2).integers(0, 10, 10).astype(np.float64)

In [119]: ts = pd.Series(data, index=idx)

In [120]: ts.iloc[[1, 2, 5, 6, 9]] = np.nan

In [121]: ts
Out[121]:
2020-01-01    8.0
2020-01-02    NaN
2020-01-03    NaN
2020-01-04    2.0
2020-01-05    4.0
2020-01-06    NaN
2020-01-07    NaN
2020-01-08    0.0
2020-01-09    3.0
2020-01-10    NaN
Freq: D, dtype: float64

In [122]: ts.plot()
Out[122]: <Axes: >
```



```
In [123]: ts.interpolate()
Out[123]:
2020-01-01    8.000000
2020-01-02    6.000000
2020-01-03    4.000000
2020-01-04    2.000000
2020-01-05    4.000000
2020-01-06    2.666667
2020-01-07    1.333333
2020-01-08    0.000000
2020-01-09    3.000000
2020-01-10    3.000000
Freq: D, dtype: float64

In [124]: ts.interpolate().plot()
Out[124]: <Axes: >
```



Interpolation relative to a `Timestamp` in the `DatetimeIndex` is available by setting `method="time"`

```
In [125]: ts2 = ts.iloc[[0, 1, 3, 7, 9]]

In [126]: ts2
Out[126]:
2020-01-01    8.0
2020-01-02    NaN
2020-01-04    2.0
2020-01-08    0.0
2020-01-10    NaN
dtype: float64

In [127]: ts2.interpolate()
Out[127]:
2020-01-01    8.0
2020-01-02    5.0
2020-01-04    2.0
2020-01-08    0.0
2020-01-10    0.0
dtype: float64

In [128]: ts2.interpolate(method="time")
Out[128]:
2020-01-01    8.0
2020-01-02    6.0
2020-01-04    2.0
2020-01-08    0.0
2020-01-10    0.0
dtype: float64
```

For a floating-point index, use `method='values'`:

```
In [129]: idx = [0.0, 1.0, 10.0]

In [130]: ser = pd.Series([0.0, np.nan, 10.0], idx)

In [131]: ser
Out[131]:
0.0      0.0
1.0      NaN
10.0    10.0
dtype: float64

In [132]: ser.interpolate()
Out[132]:
0.0      0.0
1.0      5.0
10.0    10.0
dtype: float64

In [133]: ser.interpolate(method="values")
Out[133]:
0.0      0.0
1.0      1.0
10.0    10.0
dtype: float64
```

If you have scipy installed, you can pass the name of a 1-d interpolation routine to `method`. as specified in the scipy interpolation documentation and reference guide. The appropriate interpolation method will depend on the data type.

Tip

If you are dealing with a time series that is growing at an increasing rate, use `method='barycentric'`.

If you have values approximating a cumulative distribution function, use `method='pchip'`.

To fill missing values with goal of smooth plotting use `method='akima'`.

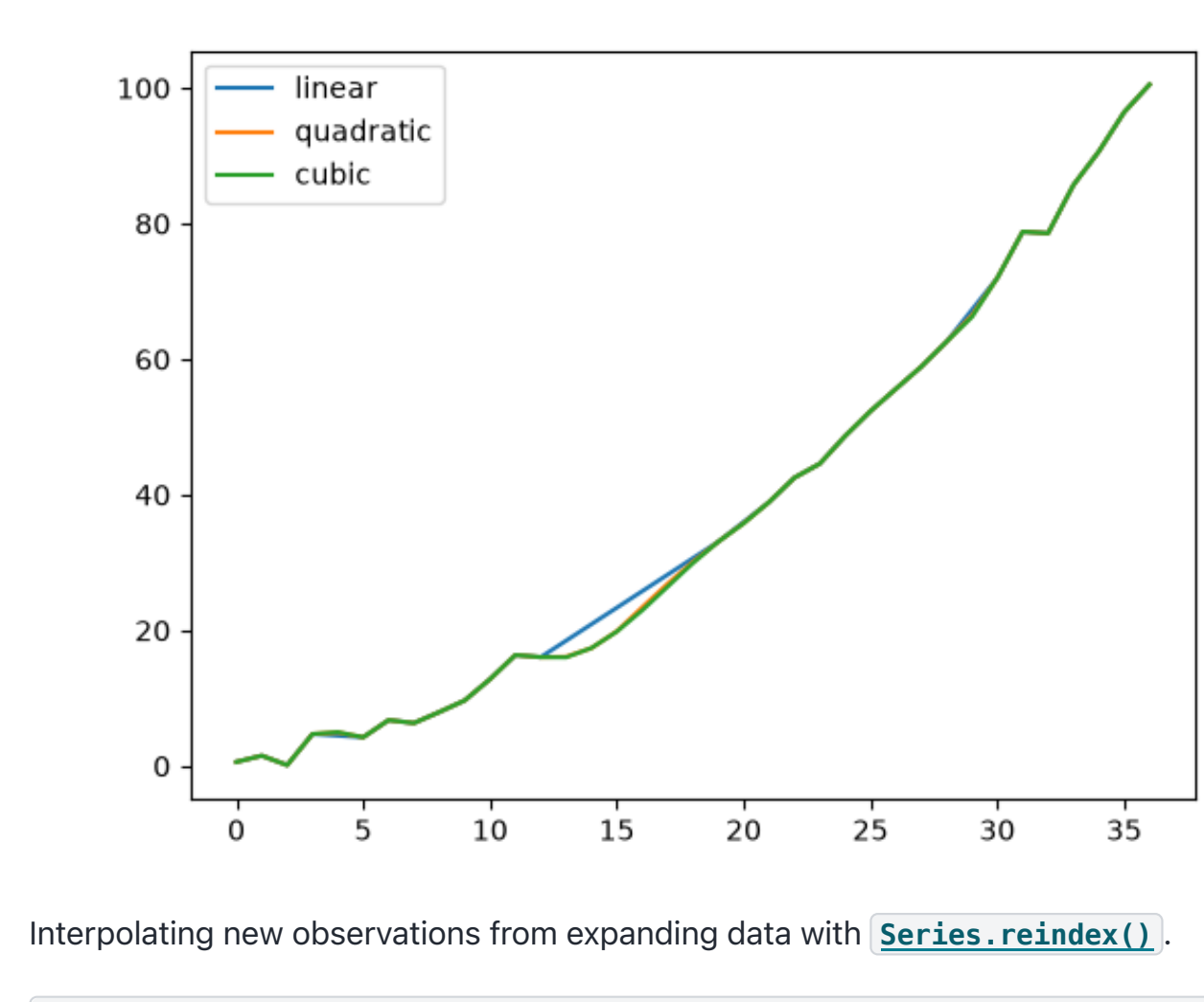
```
In [134]: df = pd.DataFrame(
   .....:     {
   .....:         "A": [1, 2.1, np.nan, 4.7, 5.6, 6.8],
   .....:         "B": [0.25, np.nan, np.nan, 4, 12.2, 14.4],
   .....:     }
   .....: )
   .....:

In [135]: df
Out[135]:
     A      B
0  1.0   0.25
1  2.1    NaN
2  NaN    NaN
3  4.7   4.00
4  5.6  12.20
5  6.8  14.40

In [136]: df.interpolate(method="barycentric")
Out[136]:
      A       B
0  1.00   0.250
1  2.10  -7.660
2  3.53  -4.515
3  4.70   4.000
4  5.60  12.200
5  6.80  14.400

In [137]: df.interpolate(method="pchip")
Out[137]:
         A          B
0  1.00000   0.250000
1  2.10000   0.672808
2  3.43454   1.928950
3  4.70000   4.000000
4  5.60000  12.200000
5  6.80000  14.400000

In [138]: df.interpolate(method="akima")
Out[138]:
          A          B
0  1.000000   0.250000
1  2.100000  -0.873316
2  3.406667   0.320034
3  4.700000   4.000000
4  5.600000  12.200000
5  6.800000  14.400000
```

When interpolating via a polynomial or spline approximation, you must also specify the degree or order of the approximation:

```
In [139]: df.interpolate(method="spline", order=2)
Out[139]:
          A          B
0  1.000000   0.250000
1  2.100000  -0.428598
2  3.404545   1.206900
3  4.700000   4.000000
4  5.600000  12.200000
5  6.800000  14.400000

In [140]: df.interpolate(method="polynomial", order=2)
Out[140]:
          A          B
0  1.000000   0.250000
1  2.100000  -2.703846
2  3.451351  -1.453846
3  4.700000   4.000000
4  5.600000  12.200000
5  6.800000  14.400000
```

Comparing several methods.

```
In [141]: np.random.seed(2)

In [142]: ser = pd.Series(np.arange(1, 10.1, 0.25) ** 2 + np.random.randn(37))

In [143]: missing = np.array([4, 13, 14, 15, 16, 17, 18, 20, 29])

In [144]: ser.iloc[missing] = np.nan

In [145]: methods = ["linear", "quadratic", "cubic"]

In [146]: df = pd.DataFrame({m: ser.interpolate(method=m) for m in methods})

In [147]: df.plot()
Out[147]: <Axes: >
```



Interpolating new observations from expanding data with `Series.reindex()`.

```
In [148]: ser = pd.Series(np.sort(np.random.uniform(size=100)))

# interpolate at new_index
In [149]: new_index = ser.index.union(pd.Index([49.25, 49.5, 49.75, 50.25, 50.5, 50

In [150]: interp_s = ser.reindex(new_index).interpolate(method="pchip")

In [151]: interp_s.loc[49:51]
Out[151]:
49.00    0.471410
49.25    0.476841
49.50    0.481780
49.75    0.485998
50.00    0.489266
50.25    0.491814
50.50    0.493995
50.75    0.495763
51.00    0.497074
dtype: float64
```

Interpolation limits

`interpolate()` accepts a `limit` keyword argument to limit the number of consecutive `NaN` values filled since the last valid observation

```
In [152]: ser = pd.Series([np.nan, np.nan, 5, np.nan, np.nan, np.nan, 13, np.nan, n

In [153]: ser
Out[153]:
0     NaN
1     NaN
2     5.0
3     NaN
4     NaN
5     NaN
6    13.0
7     NaN
8     NaN
dtype: float64

In [154]: ser.interpolate()
Out[154]:
0     NaN
1     NaN
2     5.0
3     7.0
4     9.0
5    11.0
6    13.0
7    13.0
8    13.0
dtype: float64

In [155]: ser.interpolate(limit=1)
Out[155]:
0     NaN
1     NaN
2     5.0
3     7.0
4     NaN
5     NaN
6    13.0
7    13.0
8     NaN
dtype: float64
```

By default, `NaN` values are filled in a `forward` direction. Use `limit_direction` parameter to fill `backward` or from `both` directions.

```
In [156]: ser.interpolate(limit=1, limit_direction="backward")
Out[156]:
0     NaN
1     5.0
2     5.0
3     NaN
4     NaN
5    11.0
6    13.0
7     NaN
8     NaN
dtype: float64

In [157]: ser.interpolate(limit=1, limit_direction="both")
Out[157]:
0     NaN
1     5.0
2     5.0
3     7.0
4     NaN
5    11.0
6    13.0
7    13.0
8     NaN
dtype: float64

In [158]: ser.interpolate(limit_direction="both")
Out[158]:
0     5.0
1     5.0
2     5.0
3     7.0
4     9.0
5    11.0
6    13.0
7    13.0
8    13.0
dtype: float64
```

By default, `NaN` values are filled whether they are surrounded by existing valid values or outside existing valid values. The `limit_area` parameter restricts filling to either inside or outside values.

```
# fill one consecutive inside value in both directions
In [159]: ser.interpolate(limit_direction="both", limit_area="inside", limit=1)
Out[159]:
0     NaN
1     NaN
2     5.0
3     7.0
4     NaN
5    11.0
6    13.0
7     NaN
8     NaN
dtype: float64

# fill all consecutive outside values backward
In [160]: ser.interpolate(limit_direction="backward", limit_area="outside")
Out[160]:
0     5.0
1     5.0
2     5.0
3     NaN
4     NaN
5     NaN
6    13.0
7     NaN
8     NaN
dtype: float64

# fill all consecutive outside values in both directions
In [161]: ser.interpolate(limit_direction="both", limit_area="outside")
Out[161]:
0     5.0
1     5.0
2     5.0
3     NaN
4     NaN
5     NaN
6    13.0
7    13.0
8    13.0
dtype: float64
```

Replacing values

`Series.replace()` and `DataFrame.replace()` can be used similar to `Series.fillna()` and `DataFrame.fillna()` to replace or insert missing values.

```
In [162]: df = pd.DataFrame(np.eye(3))

In [163]: df
Out[163]:
     0    1    2
0  1.0  0.0  0.0
1  0.0  1.0  0.0
2  0.0  0.0  1.0

In [164]: df_missing = df.replace(0, np.nan)

In [165]: df_missing
Out[165]:
     0    1    2
0  1.0  NaN  NaN
1  NaN  1.0  NaN
2  NaN  NaN  1.0

In [166]: df_filled = df_missing.replace(np.nan, 2)

In [167]: df_filled
Out[167]:
     0    1    2
0  1.0  2.0  2.0
1  2.0  1.0  2.0
2  2.0  2.0  1.0
```

Replacing more than one value is possible by passing a list.

```
In [168]: df_filled.replace([1, 44], [2, 28])
Out[168]:
     0    1    2
0  2.0  2.0  2.0
1  2.0  2.0  2.0
2  2.0  2.0  2.0
```

Replacing using a mapping dict.

```
In [169]: df_filled.replace({1: 44, 2: 28})
Out[169]:
      0     1     2
0  44.0  28.0  28.0
1  28.0  44.0  28.0
2  28.0  28.0  44.0
```

Regular expression replacement

Note

Python strings prefixed with the `r` character such as `r'hello world'` are "raw" strings. They have different semantics regarding backslashes than strings without this prefix. Backslashes in raw strings will be interpreted as an escaped backslash, e.g., `r'\' == '\\'`.

Replace the '.' with `NaN`

```
In [170]: d = {"a": list(range(4)), "b": list("ab.."), "c": ["a", "b", np.nan, "d"]

In [171]: df = pd.DataFrame(d)

In [172]: df.replace(".", np.nan)
Out[172]:
   a    b    c
0  0    a    a
1  1    b    b
2  2  NaN  NaN
3  3  NaN    d
```

Replace the '.' with `NaN` with regular expression that removes surrounding whitespace

```
In [173]: df.replace(r"\s*\.\s*", np.nan, regex=True)
Out[173]:
   a    b    c
0  0    a    a
1  1    b    b
2  2  NaN  NaN
3  3  NaN    d
```

Replace with a list of regexes.

```
In [174]: df.replace([r"\.", r"(a)"], ["dot", r"\1stuff"], regex=True)
Out[174]:
   a       b       c
0  0  astuff  astuff
1  1       b       b
2  2     dot     NaN
3  3     dot       d
```

Replace with a regex in a mapping dict.

```
In [175]: df.replace({"b": r"\s*\.\s*"}, {"b": np.nan}, regex=True)
Out[175]:
   a    b    c
0  0    a    a
1  1    b    b
2  2  NaN  NaN
3  3  NaN    d
```

Pass nested dictionaries of regular expressions that use the `regex` keyword.

```
In [176]: df.replace({"b": {"b": r""}}, regex=True)
Out[176]:
   a  b    c
0  0  a    a
1  1       b
2  2  .  NaN
3  3  .    d

In [177]: df.replace(regex={"b": {r"\s*\.\s*": np.nan}})
Out[177]:
   a    b    c
0  0    a    a
1  1    b    b
2  2  NaN  NaN
3  3  NaN    d

In [178]: df.replace({"b": r"\s*(\.)\s*"}, {"b": r"\1ty"}, regex=True)
Out[178]:
   a    b    c
0  0    a    a
1  1    b    b
2  2  .ty  NaN
3  3  .ty    d
```

Pass a list of regular expressions that will replace matches with a scalar.

```
In [179]: df.replace([r"\s*\.\s*", r"a|b"], "placeholder", regex=True)
Out[179]:
   a            b            c
0  0  placeholder  placeholder
1  1  placeholder  placeholder
2  2  placeholder          NaN
3  3  placeholder            d
```

All of the regular expression examples can also be passed with the `to_replace` argument as the `regex` argument. In this case the `value` argument must be passed explicitly by name or `regex` must be a nested dictionary.

```
In [180]: df.replace(regex=[r"\s*\.\s*", r"a|b"], value="placeholder")
Out[180]:
   a            b            c
0  0  placeholder  placeholder
1  1  placeholder  placeholder
2  2  placeholder          NaN
3  3  placeholder            d
```

Note

A regular expression object from `re.compile` is a valid input as well.

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